The minimum degree of nonnegative multiples of cyclotomic polynomials

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Abstract

Let $\Phi_n(x)$ be the n-th cyclotomic polynomial, let p be the smallest prime divisor of $n > 1$, and put $P = n/p$. Steinberger formulated the conjecture that the lowest-degree monic polynomial with nonnegative real coefficients divisible by Φ_n is

$$R_{n,p}(x) = 1 + x^P + \cdots + x^{(p-1)P}.$$

We prove this conjecture for every $n > 1$. More precisely, every nonzero polynomial $f \in \mathbb{R}_{\geq 0}[x]$ divisible by Φ_n has degree at least $(p-1)P$, and equality holds exactly for the positive scalar multiples of $R_{n,p}$.

The proof uses an explicit trigonometric separation certificate involving only the primitive frequencies $1, \dots, p-1$. With $\alpha = \pi/p$, define

$$K_p(\theta) = \sum_{k=1}^{p-1} \sin(k\alpha)\cos(k\theta).$$

An elementary closed form shows that the shifted values $K_p(2\pi j/n + \alpha)$ vanish at the regular p-gon positions and have one strict sign at every other exponent below the proposed degree. Evaluation at primitive n-th roots makes this vector orthogonal to every Φ_n-multiple of degree less than n. A separate unit-group reduction argument gives uniqueness on the boundary. The argument is uniform: it does not require n to be squarefree and includes even integers, primes, prime powers, and integers with arbitrarily many prime factors.

1 Introduction

For an integer $n > 1$, define

$$\mu(n) = \min\{\deg f : 0 \neq f \in \mathbb{R}_{\geq 0}[x],\ \Phi_n(x) \mid f(x)\}. \tag{1.1}$$

If p is the smallest prime divisor of n and $P = n/p$, then

$$R_{n,p}(x) = 1 + x^P + \cdots + x^{(p-1)P} = \Phi_p(x^P) \tag{1.2}$$

vanishes at every primitive n-th root of unity. Hence $\Phi_n \mid R_{n,p}$, and

$$\mu(n) \leq (p-1)P. \tag{1.3}$$

Steinberger [1, Conjecture 1] conjectured equality in (1.3), with $R_{n,p}$ as the unique monic minimizer. He proved the conjecture when n is even, when n is a prime power, and, writing the other distinct prime divisors as $q_1, \dots, q_k$, whenever

$$\frac{2}{p} > \frac{1}{q_1} + \cdots + \frac{1}{q_k}. \tag{1.4}$$

The same paper explicitly expressed disbelief in the full conjecture and anticipated counterexamples for orders with many large prime factors. Its general method reduces the squarefree case to a finite separation problem in a high-dimensional zero-sum array space.

The purpose of this paper is to give a uniform affirmative solution. The certificate below is not obtained by solving those high-dimensional systems. It lies in a $(p-1)$-frequency subspace and is available in closed form for every n.

Theorem 1.1. *Let $n > 1$, let p be its smallest prime divisor, and put $P = n/p$. If $0 \neq f \in \mathbb{R}_{\geq 0}[x]$ and $\Phi_n \mid f$, then*

$$\deg f \geq (p-1)P. \tag{1.5}$$

Equality holds if and only if

$$f(x) = cR_{n,p}(x) \tag{1.6}$$

for some $c > 0$. Consequently $R_{n,p}$ is the unique monic minimizer.

The proof is self-contained. Section 2 establishes the trigonometric kernel identity and its exact sign pattern. Section 3 turns the kernel into a separation certificate for every order. Section 4 proves the boundary rigidity needed for uniqueness, including nonsquarefree orders. Section 5 proves Theorem 1.1. Section 6 records exact destructive checks, and Section 7 positions the result relative to earlier work.

2 A sector kernel

Fix a prime p, and put

$$\alpha = \frac{\pi}{p}, \qquad K_p(\theta) = \sum_{k=1}^{p-1} \sin(k\alpha)\cos(k\theta). \tag{2.1}$$

Angles are understood modulo 2π. The open cyclic sector $(-\alpha, \alpha)$ means the arc represented by real angles strictly between $-\alpha$ and α.

Lemma 2.1. *If $\theta \not\equiv \pm\alpha \pmod{2\pi}$, then*

$$K_p(\theta) = \frac{\sin\alpha\,[1 + \cos(p\theta)]}{2(\cos\theta - \cos\alpha)}. \tag{2.2}$$

The zeros of K_p are exactly the odd multiples of α. Moreover, $K_p(\theta) > 0$ for $\theta \in (-\alpha, \alpha)$, and $K_p(\theta) < 0$ at every point outside this sector that is not an odd multiple of α modulo 2π.

Proof. Set

$$S(z) = \sum_{k=1}^{p-1} \sin(k\alpha) z^k.$$

The recurrence

$$\sin((k+1)\alpha) - 2\cos\alpha \sin(k\alpha) + \sin((k-1)\alpha) = 0$$

together with $\sin(p\alpha) = 0$ and $\sin((p-1)\alpha) = \sin\alpha$ gives

$$(1 - 2z\cos\alpha + z^2)S(z) = \sin\alpha\,(z + z^{p+1}). \tag{2.3}$$

Substituting $z = e^{i\theta}$, dividing away from $\theta \equiv \pm\alpha$, and taking real parts yields (2.2). At the two removable points, (2.1) gives

$$K_p(\pm\alpha) = \frac{1}{2}\sum_{k=1}^{p-1} \sin(2k\alpha) = 0. \tag{2.4}$$

Away from these points, the numerator in (2.2) is zero exactly when $p\theta \equiv \pi \pmod{2\pi}$, hence exactly at the remaining odd multiples of α. At all other points the numerator is positive. The denominator has the sign of $\cos\theta - \cos\alpha$, which is positive exactly in $(-\alpha, \alpha)$. This proves the zero set and sign statement. For $p = 2$, the same calculation reduces to $K_2(\theta) = \cos\theta$. $\square$

3 A uniform separation certificate

Return to an arbitrary integer $n > 1$. Let p be its smallest prime divisor, put $P = n/p$, and define

$$d = (p-1)P, \qquad H = \{0, P, 2P, \ldots, (p-1)P\}. \tag{3.1}$$

For $0 \le j < n$, set

$$w_j = K_p\left(\frac{2\pi j}{n} + \alpha\right). \tag{3.2}$$

Lemma 3.1. *The vector $w = (w_0, \ldots, w_{n-1})$ satisfies*

$$w_j = 0 \quad (j \in H), \qquad w_j < 0 \quad (0 \le j < d,\ j \notin H). \tag{3.3}$$

Proof. If $j = hP$, then the angle in (3.2) is $(2h+1)\alpha$, so Lemma 2.1 gives $w_j = 0$. Conversely, an odd-multiple zero in (3.2) implies $j \equiv hP \pmod{n}$, and the representatives in $[0, n)$ are precisely the elements of H.

The angle in (3.2) lies in the positive cyclic sector precisely when

$$1 - \frac{1}{p} < \frac{j}{n} < 1,$$

or equivalently $d < j < n$. Thus every nonanchor with $0 \le j < d$ lies in the strict negative region of Lemma 2.1. If $P = 1$, the strict-negative index set is empty. If $p = 2$, (3.2) becomes $w_j = -\sin(2\pi j/n)$, giving the same conclusion directly. $\square$

Lemma 3.2. *Let $f(x) = \sum_{j=0}^{n-1} a_j x^j \in \mathbb{R}[x]$ and suppose $\Phi_n \mid f$. Then*

$$\sum_{j=0}^{n-1} a_j w_j = 0. \tag{3.4}$$

Proof. Let $\zeta_n = e^{2\pi i/n}$. For every integer k with $1 \le k < p$, every prime divisor of k is smaller than p and therefore cannot divide n. Hence $(k, n) = 1$, so ζ_n^k is a primitive n-th root and

$$f(\zeta_n^k) = 0. \tag{3.5}$$

Using (2.1) and (3.2), we obtain

$$\sum_{j=0}^{n-1} a_j w_j = \sum_{k=1}^{p-1} \sin(k\alpha) \sum_{j=0}^{n-1} a_j \cos\Big(\frac{2\pi kj}{n} + k\alpha\Big)$$
$$= \sum_{k=1}^{p-1} \sin(k\alpha)\ \mathrm{Re}\Big(e^{ik\alpha} f(\zeta_n^k)\Big) = 0. \tag{3.6}$$

No squarefreeness or Chinese-remainder decomposition is used. □

4 Boundary rigidity for arbitrary orders

The separation certificate will force a minimizer onto H. We next show that divisibility then forces all anchor coefficients to be equal.

Lemma 4.1. *The reduction homomorphism*

$$(\mathbb{Z}/n\mathbb{Z})^* \longrightarrow (\mathbb{Z}/p\mathbb{Z})^* \tag{4.1}$$

is surjective.

Proof. Write $n = p^e m$, where $e \geq 1$ and $(p, m) = 1$. Given $b \in (\mathbb{Z}/p\mathbb{Z})^*$, choose a unit b' modulo p^e reducing to b. If $m > 1$, the Chinese remainder theorem gives an integer a satisfying

$$a \equiv b' \pmod{p^e}, \qquad a \equiv 1 \pmod{m}.$$

Then $(a, n) = 1$ and $a \equiv b \pmod{p}$. The case $m = 1$ is immediate. □

Lemma 4.2. *Suppose* $0 \neq f \in \mathbb{R}_{\geq 0}[x]$*,* $\Phi_n \mid f$*,* $\deg f \leq d$*, and* $\mathrm{supp}(f) \subseteq H$*. Then*

$$f(x) = cR_{n,p}(x) \tag{4.2}$$

for some $c > 0$*.*

Proof. Write $f(x) = g(x^P)$, where $\deg g \leq p - 1$. If ζ_n is a primitive n-th root, then $\zeta_p = \zeta_n^P$ has order p. For every $a \in (\mathbb{Z}/n\mathbb{Z})^*$,

$$0 = f(\zeta_n^a) = g(\zeta_p^a). \tag{4.3}$$

By Lemma 4.1, the exponents $a \bmod p$ range over all nonzero residue classes. Therefore g vanishes at every primitive p-th root. Hence $\Phi_p \mid g$. Since $g \neq 0$, $\deg g \leq p - 1 = \deg \Phi_p$, and $\Phi_p = 1 + x + \cdots + x^{p-1}$, we have $g = c\Phi_p$. The coefficient conditions force $c > 0$, proving (4.2). □

5 Proof of the main theorem

The polynomial $R_{n,p} = \Phi_p(x^P)$ vanishes at every primitive n-th root: if ζ_n is primitive, then ζ_n^P is a nontrivial p-th root. Thus $\Phi_n \mid R_{n,p}$, and (1.3) holds.

Conversely, suppose that $0 \neq f \in \mathbb{R}_{\geq 0}[x]$, $\Phi_n \mid f$, and $\deg f \leq d$. Because $d < n$, write

$$f(x) = \sum_{j=0}^{n-1} a_j x^j$$

by padding with zero coefficients. Lemma 3.2 gives

$$0 = \sum_{j=0}^{d} a_j w_j. \tag{5.1}$$

Every term in (5.1) is nonpositive by Lemma 3.1. At each nonanchor $0 \le j < d$, it is strictly negative if $a_j > 0$. Therefore all such coefficients vanish and $\operatorname{supp}(f) \subseteq H$. Lemma 4.2 now gives $f = cR_{n,p}$ with $c > 0$.

It follows simultaneously that no nonzero admissible polynomial has degree less than d, and that every degree-d admissible polynomial is a positive scalar multiple of $R_{n,p}$. This proves Theorem 1.1.

6 Exact destructive checks

The proof above is symbolic and does not depend on computation. We nevertheless used exact and high-precision checks to attack the most likely failure modes. The accompanying verifier `verification/verify_full_n_sector_certificate.py` checks:

- every $2 \le n \le 1000$, separated into primes, prime powers, mixed squarefree orders, and mixed nonsquarefree orders;
- every anchor by direct high-precision finite summation for $2 \le n \le 300$, selected anchors above that range, and every nonanchor prefix and tail sign for $2 \le n \le 1000$;
- exact gcd checks for primitivity of all frequencies $1, \dots, p-1$;
- surjectivity of $(\mathbb{Z}/n\mathbb{Z})^* \to (\mathbb{Z}/p\mathbb{Z})^*$;
- for every $2 \le n \le 180$, divisibility of $R_{n,p}$ and 6410 direct orthogonality checks on shifted Φ_n-basis elements;
- direct finite-sum versus closed-form evaluations at 24 nontrivial angles;
- a phase-removal negative control and a wrong-prime negative control.

Run it from this entry directory with

```
python -X utf8 .\verification\verify_full_n_sector_certificate.py
```

The verifier SHA-256 is

`62E512FD23DB231D05E08BDB074F9F36C37CC5FE0AC7337CA88BB0C249895AB3`.

All checks pass. Removing the phase destroys the anchor zeros, while replacing the smallest prime by an arbitrary prime can introduce nonprimitive low frequencies, as expected. These computations are destructive controls only; the all-orders conclusion rests on Lemmas 2.1–4.2.

7 Relation to earlier work and scope

Steinberger [1] introduced problem (1.1), formulated Conjecture 1, proved the reciprocal sufficient condition (1.4), and developed a squarefree array separation criterion. The present proof uses the same broad dual-separation idea, but its certificate is the explicit low-frequency vector (3.2), whose membership and sign are established directly and uniformly for every order.

Reddy [2] studies a different restricted problem involving $0, 1$-polynomials and minimal vanishing sums, principally for even orders with few distinct prime factors. Kepka and Korbelář [3] give degree bounds for nonnegative multiples of a general polynomial and solve their cubic case. Łaba and Zakharov [4] use Steinberger's bound in the study of integer tilings. None of these results states Theorem 1.1.

Two earlier public-beta notes by the present author treated portions of the same problem. The first [5] gave exact dense certificates for the first two orders outside Steinberger's theorem. The second [6] constructed symbolic tail-projection certificates for infinite prime families outside (1.4). Theorem 1.1 strictly subsumes the mathematical conclusions of both notes; their certificate constructions remain separate computational and structural records, but they are not needed for the proof here.

A targeted literature refresh on 28 August 2026 checked the original article, its current OpenAlex forward-citation set, arXiv, and exact-phrase and method-level searches. OpenAlex listed ten citing works; the newest was [4], and the only directly adjacent lowest-degree papers were [2, 3]. We found no public source claiming an all-orders proof, the sector certificate (3.2), or an affirmative resolution of Conjecture 1. This is a bounded public search, not proof of global priority: unindexed, unpublished, or concurrent work could still affect the novelty assessment.

8 AI-assisted research and writing disclosure

OpenAI Codex was used for computational exploration, proof development and organization, adversarial checking, literature search, and language editing. Carptopus selected and reviewed the retained mathematical claims and assumes responsibility for the manuscript.

References

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